

د. جودت الكساسة  
الامتحان 2023-1-30



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مادة الأنظمة والأختارات

| Question | 1.  | 2.  | 3.  | 4.  | 5.  | 6.  | 7.  | 8.  | 9.  | 10. |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Answer   | b   | a   | a   | c   | b   | b   | b   | d   | c   | b   |
| Question | 11. | 12. | 13. | 14. | 15. | 16. | 17. | 18. | 19. | 20. |
| Answer   | b   | d   | b   | a   | d   | d   | c   | d   | a   | b   |
| Question | 21. | 22. | 23. | 24. | 25. | 26. | 27. | 28. | 29. | 30. |
| Answer   | d   | a   | b   | b   | b   | b   | d   | a   | d   | c   |
| Question | 31. | 32. | 33. | 34. |     |     |     |     |     |     |
| Answer   | c   | c   |     |     |     |     |     |     |     |     |

Q1: (34 marks) choose the correct answer

1. If  $F(s) = \frac{5(s+250)}{s(s+100)}$ , So the final value of  $F(s)$  equals

a. 5

c. 0.05

b. 12.5

d. 1250

2. The inverse of Laplace form of the  $F(s) = \frac{2s+5}{s^2+5s+6}$  is

a.  $f(t) = 2e^{(-2.5t)} \cosh(0.5t)$

b.  $f(t) = e^{(-2t)} + e^{(-3t)}$

c.  $f(t) = 2e^{(-2.5t)} \sinh(0.5t)$

d.  $f(t) = 2e^{(-2.5t)} \cos(0.5t)$

3. The Laplace transform of  $f(t) = 2e^{(10t)} \cos(10t)$  is

a.  $F(s) = \frac{2(s-10)}{s^2-20s+200}$

b.  $F(s) = \frac{2(s+10)}{s^2+20s+200}$

c.  $F(s) = \frac{2(s-10)}{s^2-20s}$

d.  $F(s) = \frac{2(s-10)}{s^2+100}$

4. The Laplace transform of  $(t^n - 1)$  where  $n$  is integer is

a.  $F(S) = \frac{1}{s^n}$

b.  $F(S) = \frac{(n-1)!}{s^n}$

c.  $F(S) = \frac{1}{(n-1)s^n}$

d.  $F(S) = \frac{(n)!}{s^n}$

5. The Fourier transform of a signal  $x(t) = e^{2t}u(-t)$  is given by

a.  $\frac{1}{2-j\omega}$

b.  $\frac{2}{2-j\omega}$

c.  $\frac{1}{j2-\omega}$

d.  $\frac{2}{j2-\omega}$

6. The function which has its Fourier transform and Laplace transform unity (equal one) is

a. Unit Gaussian

b. Unit Impulse

c. Unit sinc

d. Unit step

7. An LTI system has impulse response  $h(t) = u(t-1)$ . If the input is  $x(t) = u(t-2)$  the output  $y(t)$  is given by

a.  $y(t) = tu(t)$

b.  $y(t) = tu(t-1)$

c.  $y(t) = (t-1)u(t-1)$

d.  $y(t) = (t-3)u(t-3)$

8. The output of the discrete time system  $h[n] = \{1, 3, 2, 2\}$  when the input  $x[n] = \{1, -2, 3\}$  is applied

a.  $y[n] = \{1, 1, -1, 7, 2, 6\}$

b.  $y[n] = \{1, 1, -1, 7, 4, 6\}$

c.  $y[n] = \{1, 1, -1, 7, -4, 6\}$

d.  $y[n] = \{1, 1, 1, 7, 2, 6\}$

9. The Fourier transform of the exponential  $e^{j\omega_0 t}$  is:

a. A constant

b. A rectangular

c. An impulse

d. A series of impulses

10. The Fourier transform of  $x(t)e^{j\omega_0 t}$  is

a.  $\frac{1}{2}X(f-f_0)$

b.  $X(f-f_0)$

c.  $X(f+f_0)$

d.  $\frac{1}{2}X(f-f_0) + \frac{1}{2}X(f+f_0)$

11. The Laplace transform of  $x(t) = 2e^{-5t}u(t) + e^{-4t}u(-t)$ .

a.  $X(s) = \frac{2}{s+5} + \frac{-1}{s-4}$

b.  $X(s) = \frac{2}{s+5} + \frac{1}{s+4}$

c.  $X(s) = \frac{2}{s+5} - \frac{1}{s+4}$

d.  $X(s) = \frac{-2}{s+5} - \frac{1}{s-4}$

12. The frequency response of a system with  $h(t) = \delta(t) - \delta(t-1)$ , is given by

a.  $\delta(\omega) - \delta(\omega-1)$

b.  $1 - e^{j\omega}$

c.  $u(\omega) - u(\omega-1)$

d.  $1 - e^{-j\omega}$

13. For the periodic rect signal  $x(t)$  with pulse width ( $\tau$ ) and period ( $T_0$ ), if we increase the pulse width ( $\tau$ ), that would affect the resulting spectrum by:

a. Increase the spacing between the spectral components

b. decrease the spacing between the spectral components

c. Increase the width of the resulting sinc function

d. decrease the width of the resulting sinc function

14. For the periodic rect signal  $x(t)$  with pulse width ( $\tau$ ) and period ( $T_0$ ), if we decrease the period ( $T_0$ ), that would affect the resulting spectrum by:

a. Increase the spacing between the spectral components

- b. decrease the spacing between the spectral components
- c. Increase the width of the resulting sinc function
- d. decrease the width of the resulting sinc function

15. A signal  $m(t)$  is multiplied by a sinusoidal waveform of frequency  $(fc)$  such that

$v(t) = 5m(t) \cos 2\pi f_c t$ . If Fourier transform of  $m(t)$  is  $M(f)$ , Fourier transform of  $v(t)$  will be

- a.  $2.5 M(f + 2f_c)$
- b.  $5 M(f - f_c)$
- c.  $5 M(f + f_c) + 5 M(f - f_c)$
- d.  $2.5 M(f + 2f_c) + 2.5 M(f - 2f_c)$

16. The impulse response  $h(n)$  of a linear time invariant system is given by:

$h(n) = u(n+3) + u(n-2) + 2u(n-7)$ , where  $u(n)$  is the unit step sequence. The above system is:

- a. stable but not causal
- b. stable and causal
- c. causal but unstable
- d. unstable and not causal

17. Discrete time signal is derived from continuous time signal by \_\_\_\_\_ process.

- a. Addition
- b. Multiplying
- c. Sampling
- d. convolution

18. The condition for memory-less system is given by \_\_\_\_\_

- a.  $h[k] = u[k]$
- b.  $h[k] = u[n-k]$
- c.  $h[k] = h[k]u[k]$
- d.  $h[k] = h[n-k]u[k]$

19. Which is the commutative property of the LTI System in case of discrete time system?

- a.  $x[n] + h[n] = h[n] + x[n]$
- b.  $x[n] + h[n] = h[n] * x[n]$
- c.  $x[n] * h[n] = h[n] * x[n]$
- d.  $x[t] * h[t] = h[n] * x[n]$

20. Discrete-time signals are \_\_\_\_\_

- a. Continuous in amplitude and continuous in time
- b. Continuous in amplitude and discrete in time
- c. Discrete in amplitude and discrete in time
- d. Discrete in amplitude and continuous in time

21. For a continuous-time signal  $x(t)$  to be periodic with a period  $T$ , then  $x(t + mT)$  should be equal to:

- a.  $x(-t)$
- b.  $x(mT)$
- c.  $x(mt)$
- d.  $x(t)$

22. For the exponential signal  $e^{-j500\pi t}$ , the fundamental frequency and period equal to:

- a.  $\omega_0 = 500\pi$ ,  $T_0 = 4$  ms
- b.  $\omega_0 = 500\pi$ ,  $T_0 = -4$  ms
- c.  $\omega_0 = -500\pi$ ,  $T_0 = 4$  ms
- d.  $\omega_0 = -500\pi$ ,  $T_0 = -4$  ms

23. The energy and power of the periodic complex exponential signal  $e^{-j\omega_0 t}$  are:

- a. Energy is  $\infty$ , power is 1.
- b. Energy is  $\infty$ , power is  $T_0$
- c. Energy is  $T_0$ , power is 1.
- d. Energy is  $\omega_0$ , power is 1

24. The Fourier transform of the signal  $x(t) = e^{3t} u(-t)$  is

a.  $X(j\omega) = \frac{1}{3+j\omega}$

b.  $X(j\omega) = \frac{1}{3-j\omega}$

c.  $X(j\omega) = \frac{6}{3+j\omega^2}$

d.  $X(j\omega) = \frac{3}{3+j\omega^2}$

25. If  $a_k$  are the Fourier series coefficients of a signal  $g(t)$ , the Fourier series coefficients of the signal  $dx(t)/dt$  are:

a.  $\frac{a_k}{jk\omega_0}$

b.  $jk\omega_0 a_k$

c.  $\frac{-a_k}{jk\omega_0}$

d.  $-jk\omega_0 a_k$

26. The function  $X(j\omega) = \text{sinc}(3.5\omega)$  is the Fourier transform of the signal:

a.  $x(t) = \frac{1}{7} \text{rect}(\frac{t}{7})$

b.  $x(t) = \frac{1}{3.5} \text{rect}(\frac{t}{3.5})$

c.  $x(t) = 7 \text{rect}(\frac{t}{7})$

d.  $x(t) = 3.5 \text{rect}(\frac{t}{3.5})$

27. The function  $X(j\omega) = \pi e^{|\omega|}$  is the Fourier transform of the signal:

a.  $x(t) = \frac{2}{1+t^2}$

b.  $x(t) = \frac{1}{1+t^2}$

c.  $x(t) = \frac{2}{1-t^2}$

d.  $x(t) = \frac{1}{1-t^2}$

28. Using the integration property to find the Fourier transform of  $x(t)$

$$x(t) = t, \text{ for } -1 \leq t \leq 1.$$

a.  $X(j\omega) = \frac{1}{j\omega} Y(j\omega) + \pi Y(0) \delta(\omega)$

b.  $X(j\omega) = \frac{1}{j\omega} Y(j\omega) + \pi X(0) \delta(\omega)$

c.  $X(j\omega) = \frac{1}{j\omega} Y(j\omega)$

d.  $X(j\omega) = \frac{1}{j\omega} Y(j\omega) + \pi \delta(\omega)$

29. Which of the following cannot be the Fourier series expansion of a periodic signals?

a.  $x(t) = 2\cos t + 3\cos 3t$

b.  $x(t) = 2\cos \pi t + 7\cos t$

c.  $x(t) = \cos t + 0.5$

d.  $x(t) = 2\cos 1.5\pi t + \sin 3.5\pi t$

30. The trigonometric Fourier series of an even function of time does not have

a. the dc term

b. cosine terms

c. sine terms

d. odd harmonic terms

31. Choose the function  $f(t)$ ;  $-\infty < t < +\infty$ , for which a Fourier Series cannot be defined

a.  $3 \sin(25t)$

b.  $4 \cos(20t + 3) + 3 \sin(10t)$

c.  $\exp(-|t|) \sin(25t)$

d. 1

32. Which one of the following is not a ramp function?

a.  $r(t) = t$  when  $t \geq 0$

b.  $r(t) = 0$  when  $t < 0$